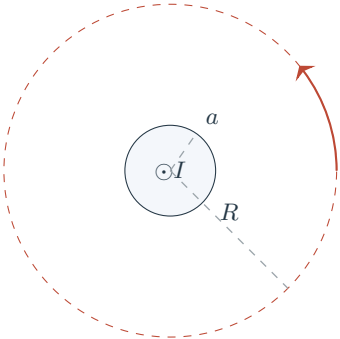
# M5-132 孤立长直导线的自感为何发散




$C_R$ 为外部场能的积分截断边界

$$W'_m = \mu_0 I^2 \ln(R/a)/(4\pi)$$

$$L'_{\text{ext}} = \mu_0 \ln(R/a)/(2\pi)$$

$$R \to \infty : \quad L'_{\text{ext}} \to \infty$$